THE INTERNAL-HOM EVALUATE MAP UNDER BOUNDED TRUNCATION HYPOTHESES ON K

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Allowed standard prerequisites (editorial).

- The canonical truncation triangle of K gives a long exact sequence. invoked at proof lines 28634-28639. Allows proving the map on truncations and controlling the rest by vanishing.
- The evaluate map is an isomorphism for pseudo-coherent truncated objects by the previous technical lemma. invoked at proof lines 28640-28641. Handles the tau_leq_n K part.
- Finite tor dimension of RHom(L, M) and finite injective dimension of L, M bound the cohomology of both sides on the upper truncations. used at proof lines 28642-28649. Finishes the induction over n.

Source reference. Original lemma 0A67: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18732–18732. Cited at statement lines 28623-28623 as the map whose isomorphism is claimed; the proof is inside the counted span.

Source reference. Original remark 08J5: sources/stacks-chow/tags.txt, source lines 10838–10838. Cited at proof lines 28638-28639 for the truncation triangle; the reduction argument is inside the counted span.

Source reference. Original lemma 0A68: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28562–28562. Cited at proof lines 28640-28641 for the isomorphism on the lower truncations; the vanishing argument and conclusion are inside the counted span.

Lemma 0.2. *Let R be a ring. Let K, L, M be objects of $D(R)$. the map*

$$R\operatorname{Hom}_R(L, M) \otimes^{\mathbf{L}}_R K \longrightarrow R\operatorname{Hom}_R(R\operatorname{Hom}_R(K, L), M)$$

of Lemma 0A67 is an isomorphism if the following three conditions are satisfied

(1) *L, M have finite injective dimension,*
(2) *$R\operatorname{Hom}_R(L, M)$ has finite tor dimension,*
(3) *for every $n \in \mathbf{Z}$ the truncation $\tau_{\leq n}K$ is pseudo-coherent*

Proof. Pick an integer n and consider the distinguished triangle

$$\tau_{\leq n}K \to K \to \tau_{\geq n+1}K \to \tau_{\leq n}K[1]$$

see Derived Categories, Remark 08J5. By assumption (3) and Lemma 0A68 the map is an isomorphism for $\tau_{\leq n}K$. Hence it suffices to show that both

$$R\operatorname{Hom}_R(L, M) \otimes^{\mathbf{L}}_R \tau_{\geq n+1}K \quad \text{and} \quad R\operatorname{Hom}_R(R\operatorname{Hom}_R(\tau_{\geq n+1}K, L), M)$$

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have vanishing cohomology in degrees $\leq n - c$ for some c. This follows immediately from assumptions (2) and (1). $\square$